

(12) **United States Plant Patent**
Jensen

(10) **Patent No.:** **US PP21,147 P2**
(45) **Date of Patent:** **Jul. 6, 2010**

- (54) **ASTER PLANT NAMED ‘DASKAT’**
- (50) Latin Name: *Aster novi-belgi*
Varietal Denomination: **DASKAT**
- (75) Inventor: **Bent Juhl Jensen**, Malling (DK)
- (73) Assignee: **Gartneriet Raahoj**, Malling (DK)
- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.
- (21) Appl. No.: **12/380,349**
- (22) Filed: **Feb. 26, 2009**
- (51) **Int. Cl.**
A01H 5/00 (2006.01)
- (52) **U.S. Cl.** **Plt./355**
- (58) **Field of Classification Search** **Plt./355**

See application file for complete search history.

- (56) **References Cited**
- OTHER PUBLICATIONS
- UPOV-ROM GTITM, Plant Variety Database, 2009/05, GTI Jouve Retrieval Software, citation for ‘DASKAT’.*
- Print-out of application number and filing date from Community Plant Variety Office (CPVO) website for corresponding, CPVO application No. 2007/2427 filed Nov. 5, 2007, together with copy of Assignment filed Jan. 23, 2009 (8 pages). (<http://www.cpvoextranet.cpvo.europa.eu>).
- * cited by examiner
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- (57) **ABSTRACT**
- A new distinct cultivar of *Aster* plant named ‘DASKAT’, characterized by its upright and inverted conical plant habit; gray-green foliage; freely flowering habit; decorative, composite-type inflorescence with about 250 to 30 red-purple-colored ray florets; and only about 10 to 15 green-colored disc florets per inflorescence (almost 100% filled capitulae).

3 Drawing Sheets

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Latin name of the genus and species of the claimed plant: *Aster novi-belgi* (now *Symphyotrichum novi-belgii*).
Variety denomination: ‘DASKAT’.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct cultivar of *Aster* plant, botanically known as *Aster novi-belgii* (now *Symphyotrichum novi-belgii*) of the Asteraceae family, commonly known as Michaelmas Daisy and New York Aster, and hereinafter referred to by the cultivar name ‘DASKAT’.

The new *Aster* cultivar is a product of a planned breeding program conducted by the inventor, Bent Juhl JENSEN, in Malling, Denmark. The objective of the breeding program is to develop a new *Aster* variety with uniform plant growth habit, unique floret colors, and good postproduction longevity.

The new *Aster* cultivar originated from a cross made in a controlled breeding program by the inventor on Apr. 1, 2005, in Malling, Denmark. The female or seed parent is *Aster novi-belgii* ‘VICTORIA MATHILDE’ (unpatented, disclosed in pending CPVO Application No. 2006/1898). The male or pollen parent is the unpatented *Aster novi-belgii* seedling selection designated 05-008-007. The new *Aster* ‘DASKAT’ was discovered and selected by the inventor as a single flowering plant within the progeny of the stated cross in September of 2006 in a controlled environment in Malling, Denmark. The selection of the new *Aster* ‘DASKAT’ was based on its uniform plant growth habit and desirable inflorescence form and ray floret color.

Asexual reproduction of the new *Aster* cultivar by vegetative tip cuttings was first performed in June of 2006 in Malling, Denmark, and has demonstrated that the combination of characteristics as herein disclosed for the new cultivar are firmly fixed and retained through successive generations of

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asexual reproduction. The new cultivar reproduces true to type.

BRIEF SUMMARY OF THE INVENTION

The following traits have been repeatedly observed and are determined to be unique characteristics of ‘DASKAT’, which in combination distinguish this *Aster* as a new and distinct cultivar:

1. Upright and inverted conical plant habit;
2. Gray-green foliage;
3. Freely flowering habit;
4. Decorative, composite-type inflorescence with about 250 to 300 red-purple-colored ray florets; and
5. Only about 10 to 15 green-colored disc florets per inflorescence (almost 100% filled capitulae).

Plants of the new *Aster* ‘DASKAT’ differ from plants of the female parent, *Aster novi-belgii* ‘VICTORIA MATHILDE’ (unpatented, disclosed in pending CPVO Application No. 2006/1898) in the characteristics described below:

1. Plants of ‘DASKAT’ are more compact than plants of ‘VICTORIA MATHILDE’;
2. Plants of ‘DASKAT’ produce more ray florets than plants of ‘VICTORIA MATHILDE’; and
3. Plants of ‘DASKAT’ produce red-purple (RHS N74B) ray florets whereas plants of ‘VICTORIA MATHILDE’ produce red-purple (RHS N66A) ray florets.

Plants of the male or pollen parent, the unpatented *Aster novi-belgii* seedling selection designated 05-008-007, are unavailable to provide a botanical comparison to plants of the new *Aster* ‘DASKAT’.

Of the many commercial cultivars known to the present inventor, the most similar in comparison to the new *Aster* ‘DASKAT’ are 1) *Aster novi-belgii* ‘VICTORIA PINK FANNY’ (patented, CPVO Grant No. 14582), 2) *Aster novi-belgii* ‘MILKA CARMINE’ (patented, CPVO Grant No.

5028). Plants of the new *Aster* ‘DASKAT’ differ from plants of 1) *Aster novi-belgii* ‘VICTORIA PINK FANNY’ and 2) *Aster novi-belgii* ‘MILKA CARMINE’ in the characteristics provided in Table 1.

TABLE 1

Char-acteristic	New Cultivar ‘DASKAT’	Comparison Cultivar ‘VICTORIA PINK FANNY’ (patented)	Comparison Cultivar ‘MILKA CARMINE’ (patented)
Leaf Shape:	Gladiate to linear	Ensiform	Gladiate
Mature Leaf Color (upper surface):	Gray-green, RHS N189B	Green, RHS 137A	Green, RHS 137A
Capitulum Size:	Depth: 15 mm	Depth: 12 mm	Depth: 10 mm
Ray Florets	About 250 to 300 ray florets per inflorescence, which are oval, slightly involute in shape, with rounded, slightly re-tuse apex	About 250 ray florets per flower, which are involute, almost tubular in shape, with rounded to slightly retuse apex	About 240 ray florets per flower, which are involute in shape, with acute apex
Mature Ray Floret Color (upper surface):	Red-purple, RHS N74B	Purple, RHS 77C	Purple, RHS 75C
Disc Florets:	About 10 to 15 disc florets per inflorescence	About 50 to 60 disc florets per inflorescence	About 30 to 40 disc florets per inflorescence
Mature Disc Floret Color:	Green, RHS 137D	Yellow-Green, RHS 151B	Yellow, RHS 9C

BRIEF DESCRIPTION OF THE PHOTOGRAPHS

The accompanying photographs illustrate the overall appearance of the new *Aster* ‘DASKAT’ showing the colors as true as is reasonably possible with colored reproductions of this type. Colors in the photographs may differ slightly from the color value cited in the detailed botanical description which accurately describe the color of ‘DASKAT’.

FIG. 1 shows a top and side view perspective a typical flowering plant of ‘DASKAT’ in a 9.5 cm pot, at 10 weeks of age after planting.

FIG. 2 shows a close-up view perspective of a typical mature inflorescence of ‘DASKAT’, at 10 weeks of age after planting.

FIG. 3 shows a close-up comparison view of a typical mature inflorescence and leaf of: 1) the new *Aster* ‘DASKAT’ (referenced by breeder designation 283-12), compared to 2) the comparison cultivar *Aster* ‘VICTORIA PINK FANNY’ (referenced by PINK FANNY), and 3) the comparison cultivar *Aster* ‘MILKA CARMINE’ (referenced by MILKA, ROSA).

DETAILED BOTANICAL DESCRIPTION

The new *Aster* ‘DASKAT’ has not been observed under all possible environmental conditions. The phenotype of the new cultivar may vary with variations in environment such as temperature, light intensity, and day length without any change in the genotype of the plant.

The aforementioned photographs, together with the following observations, measurements and values describe the new *Aster* ‘DASKAT’ as grown in a glass-covered greenhouse in Malling, Denmark, under conditions which closely approximate those generally used in commercial practice. ‘DASKAT’ plants were grown in 9.5 cm pots in a heated and lighted glass-covered greenhouse with the day temperatures ranging from 18° C. to 20° C. and the night temperature averaging 18° C. ‘DASKAT’ plants are grown under long day (20 hour) photoperiodic treatments for five weeks, followed by short day (12 hour) photoperiodic treatments for five weeks. During the long day (20 hour) photoperiodic treatments, if ambient light level falls below +50 Wm², a supplementary light at +200 Wm² is applied. ‘DASKAT’ plants were pinched once, and treated twice with the growth retardant Daminozide.

Color references are made to The Royal Horticultural Society Colour Chart (R.H.S.), 4th Edition, except where general colors of ordinary significance are used. Color values were taken under daylight conditions in Malling, Denmark. The age of the ‘DASKAT’ plants shown in the photographs and described herein is about 10 weeks.

Classification:

Botanical.—*Aster novi-belgii* (now *Symphyotrichum novi-belgii*).

Parentage:

Female or seed parent.—*Aster novi-belgii* ‘VICTORIA MATHILDE’ (unpatented, disclosed in pending CPVO Application No. 2006/1898).

Male or pollen parent.—*Aster novi-belgii* seedling selection designated 05-008-007 (unpatented).

Propagation:

Type.—Vegetative tip cuttings.

Time and temperature to initiate roots.—Summer: About 12 days at 21° C. Winter: About 13 days at 21° C.

Rooting habit.—Fine, fibrous and freely branching.

Root color.—Gray-white, close to RHS 156D.

Plant description:

General appearance and form.—Herbaceous, decorative-type potted or bedding plant with upright plant habit. *Aster* inflorescences in composite heads.

Growth and branching habit.—Stems upright; inverted conical plant habit. Appropriate for 9 cm to 15 cm containers.

Growth rate/vigor.—Vigorous.

Plant height (soil level to top of plant plane).—About 17 cm.

Plant width (spread).—About 20 cm.

Crop time to produce a mature flowering plant.—After rooting, about 10 weeks are required to produce finished flowering plants in 9.5 cm pots.

Branches:

Number of branches per plant.—About 15, with 3 cuttings per pot.

Quantity of buds and inflorescence per lateral stem.—About 2 to 6, buds continue to develop when dead flowers are removed.

Branching habit.—Freely after pinching.

Length.—About 12 cm (including flowers).

Diameter.—About 3 mm.

Internode length.—About 9 mm. Range about 7 mm to 13 mm.

Strength.—Strong.

Aspect.—Upright.

Texture.—Glabrous with 3–4 longitudinal furrows.

Color.—Green, RHS 137C.

Foliage description:

Quantity per branch.—About 10 to 14.

Arrangement.—Single, alternate, petiolated. Older 5
leaves are twisted in form.

Length.—Up to 6 cm.

Width.—Up to 14 mm.

Overall shape of leaf.—Gladiolate to linear.

Shape at apex.—Acute. 10

Shape at base.—Decurrent, auriculate ½ clasping.

Margin.—Slightly crenulate.

Texture.—Glabrous, smooth, leathery.

Pubescence.—None.

Color of developing foliage.—Upper surface: Green, 15
RHS 139A. Under surface: Gray-green, RHS 189A.

Color of mature foliage.—Upper surface: Gray-green,
RHS N189B. Under surface: Gray-green, RHS 191A.

Venation pattern.—None, but prominent central abaxial
vein. 20

Venation color.—Upper surface: Gray-green, RHS
191B. Under surface: Yellow-green, RHS 146C.

Petiole.—Length: 15 mm to 22 mm. Diameter: 1 mm to
3 mm. Appearance and angle: About 30° from verti- 25
cal. Texture: Glabrous. Color: Green, RHS 137C.

Inflorescence description:

Appearance.—Terminal and axillary composite inflo-
rescences held above and beyond the foliage, with
about 250 to 300 lanceolate-shaped ray florets and
only about 10 to 15 disc florets; ray and disc florets 30
arranged acropetally on a capitulum. Inflorescences
face upright and form is capitulum.

Natural flowering season.—Under natural season con-
ditions, plants flower in late summer through autumn
in Denmark. Season can be extended by vernalization 35
and long day treatments.

Time to flower.—About 5 to 6 weeks (longevity of indi-
vidual inflorescences is dependent on temperature
and light conditions).

Postproduction longevity.—Inflorescences maintain 40
good color and substance for about 27 days on the
plant when in an indoor environment, and may keep
longer if temperatures are maintained below 20° C.
Inflorescences persistent.

Quantity of inflorescences.—About 40 to 50 buds and 45
open inflorescences per plant.

Fragrance.—Faint.

Bud.—Rate of opening: About 10 per week, for 3 weeks
after induction. Length: Up to 7 mm. Diameter: Up to
7 mm. Shape: Globular. Texture: Glabrous. Color: 50
Green, RHS 137C.

Peduncle.—Length: 10 mm to 15 mm. Diameter: 1 mm.
Appearance and angle: About 70° from vertical.
Strength: Strong. Texture: Glabrous. Color: Green,
RHS 134C.

Inflorescence.—Inflorescence height: About 6 to 8 cm.
Inflorescence diameter About 4 to 6 cm. Capitulum
height: About 1.5 cm Capitulum diameter: About 3.5
cm

Quantity of flowers (capitulae) per inflorescence.— 60
About 10 to 14.

Ray florets.—Arrangement and quantity: Imbricate,
about 250 to 300 ray florets per capitulum in 9 to 12
whorls of ray florets (depending on light and tempera-

ture conditions). Orientation: Initially upright, later
mostly horizontal. Aspect: Straight to slightly invo-
lute. Appearance: Lanceolate, but sides turn inward
(involute). Length: About 10 mm. Width: About 2
mm. Overall shape: Oval, slightly involute. Shape at
apex: Rounded, slightly retuse. Shape at base: Fused.
Margin: Entire. Texture: Upper and under surfaces:
Smooth, glabrous, silky. Pubescence: Subtended by
numerous short, about 2 mm, white hairs. Color
(when opening and fully opened): Upper surface: red-
purple, RHS N74B. Under surface: purple, RHS 76B.
Fading: Yes, gray-purple, RHS 187B.

Disc florets.—Arrangement: About 10 to 15 disc florets,
massed at center of capitulum. Length: About 2 mm.
Width: About 1 mm. Disc area diameter: About 2 mm.
Overall shape: Tubular, elongated. Shape at apex:
Acute. Shape at base: Fused to tube. Margin: Entire.
Texture: Upper and under surfaces: Scale-like, parch-
ment thin. Pubescence: Subtended by numerous
short, about 2 mm, white hairs. Color (when opening):
Upper and under surfaces: Green, RHS 137D. Color
(when fully opened): Upper and under surfaces:
Green, RHS 137D.

Phyllaries.—Quantity per inflorescence: Involucre,
about 40 to 50. Length: About 3 to 5 mm. Width:
About 1 to 2 mm. Overall shape: Lanceolate. Apex
shape: Acute. Base shape: Truncate, fused. Margin:
Entire. Texture: Hairy, setulose edges, verrucose
abaxial surface. Color (immature): Upper surface:
Green, RHS 137B. Under surface: Green, RHS 137D.
Color (mature): Upper surface: Green, RHS 137A.

Under surface: Green, RHS 137B.

Reproductive organs:

Androecium (on disc florets only).—Stamen number: 1
or 5 per floret; fused around style. Stamen length:
About 1 mm. Anther shape: Narrowly cylindrical,
somewhat fused. Anther length: About 0.3 mm.
Anther color: Yellow, RHS 8B. Pollen amount: Mod-
erate. Pollen color: Yellow, RHS 8A.

Gynoecium (on disc and ray florets).—Pistil number: 1
per floret. Pistil length: About 2 mm. Stigma shape:
Cleft. Stigma length: About 1.2 mm. Stigma color:
White, RHS N155C. Style length: About 1 mm. Style
color: White, RHS N155D. Ovary diameter: About 2
mm. Ovary color: White, RHS N155D.

Seed: None observed.

Fruit: None observed.

Disease/pest resistance: Resistance to pathogens and pests
common to *Asters* has not been observed on plants grown
under commercial greenhouse conditions.

Disease/pest susceptibility: Resistance to pathogens and
pests common to *Asters* has not been observed on plant
grown under commercial green house conditions, except
mildew after flowers wilt.

55 High temperature tolerance: Not specifically tested, but if
enough water is available, tolerant up to 35° C.

Low temperature tolerance: Not specifically tested, but if
enough water is available, tolerant to –15° C.

I claim:

1. A new and distinct cultivar of *Aster* plant named
'DASKAT', as illustrated and described herein.

FIG. 1



FIG. 2

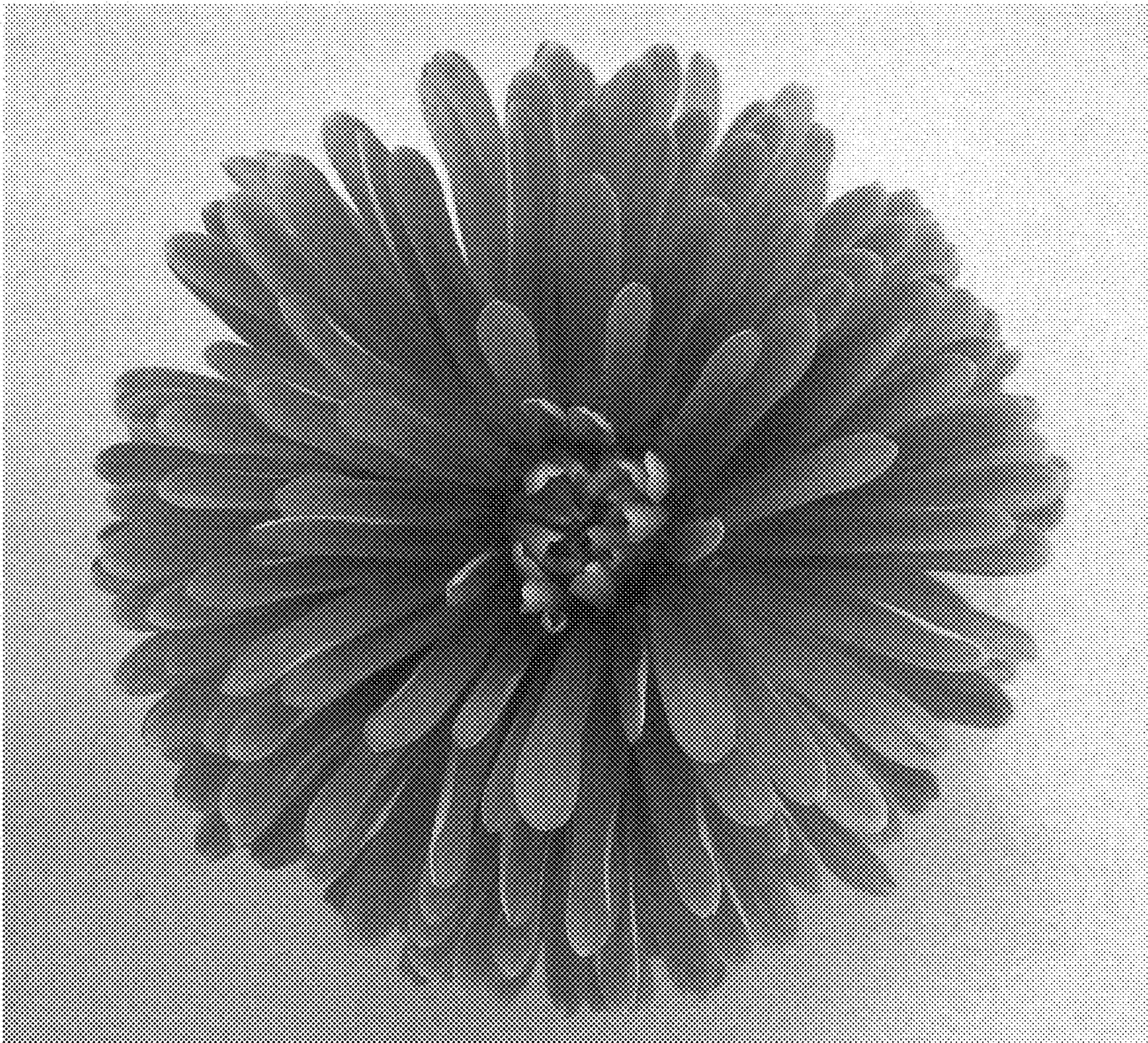


FIG. 3

